

Pre-Registered Experimental Protocol: Reward Topology and Learned Coordination

Shared-policy PPO vs Independent PPO across the alpha sweep

Core Research Document - Capstone learned-policy experiment

Decentralized Swarm Emergence Research Program

Stage One revision - September 2026

Evidence status: Stage One validates the experimental environment, instrumentation, reward accounting, heuristic baselines, and reproducibility pipeline. It does not yet establish learned emergence, a critical alpha threshold, learned communication, or a phase transition.

Protocol status

This document specifies the next paper-grade experiment before learned-policy results are interpreted. Stage One heuristic data are frozen and treated as reference evidence. Any implementation-driven protocol changes should be versioned and dated before the affected results are analyzed.

1. Primary question and hypotheses

Primary question: Does reward mixing alpha alter learned coordination under anonymous partial observability, and does policy-sharing architecture moderate that effect?

ID	Hypothesis / test
H0-A	No practically meaningful alpha-dependent change in learned global performance.
H1-A	At least one alpha regime produces a reproducible change in learned global performance.
H0-B	Shared-policy and Independent PPO do not differ materially under matched budgets.
H1-B	Policy architecture materially changes performance and/or coordination stability.
Exploratory	If supported, estimate alpha*; otherwise report no threshold.

2. Fixed environment contract

- Reference world: 16 x 16 discrete grid.
- Reference swarm: six agents and eight targets; additional sizes are secondary analyses.
- Anonymous ego-centered local observation tensor with no absolute position or identity.
- Actions: north, south, west, east, stay.
- Episode success: all targets collected before truncation.
- World generation and policy randomness recorded by seed.

3. Reward treatment

For agent i : $r_i = \alpha(\text{global_signal} - \text{step_cost}) + (1-\alpha)(\text{local_collections}_i - \text{step_cost})$. At $\alpha = 0$, collection credit is local; at $\alpha = 1$, collection credit is fully shared. The primary sweep uses alpha in $\{0.00, 0.25, 0.50, 0.75, 1.00\}$, preserving the frozen Stage One grid.

4. Learned conditions

Dimension	Primary design
Architecture	Shared-policy PPO; Independent PPO

Alpha	0, 0.25, 0.5, 0.75, 1.0
Training seeds	Minimum 10 targeted after pilot; pilot may begin with 3
Evaluation	Held-out episode seeds separated from training
Budget	Identical environment-step/training budget per condition
Baselines	Frozen random, greedy, flocking, distributed-greedy oracle

5. Outcomes

Primary outcomes: task success, completion rate, steps-to-completion, and coordination efficiency. Secondary behavioral outcomes: total distance, collision rate, action entropy, spatial order, mean inter-agent distance, role differentiation, and team return.

6. Statistical plan

- Report condition means with uncertainty intervals and seed-level distributions.
- Use effect sizes in addition to null-hypothesis significance tests.
- Treat training seed, not individual evaluation episode, as the primary independent replication unit for learned-policy comparisons.
- Correct or explicitly control family-wise/multiple-comparison interpretation when testing many alpha pairs.
- Estimate any α^* with an uncertainty procedure and a pre-specified performance/behavior criterion; do not choose it by eye.
- Inspect failure distributions and seed instability rather than reporting only the best checkpoint.

7. Emergence decision rule

A condition will not be labeled emergent merely because reward or success improves. A defensible emergence claim requires: (1) reproducible improvement in a global task measure relative to appropriate controls; (2) structured coordination in at least one pre-specified behavioral metric or trajectory analysis; and (3) evidence that the behavior is not directly encoded by the policy architecture, oracle assignment, or reward shaping. If these conditions are not met, the result will be described more narrowly as learned performance or coordination.

8. Stopping and exclusion rules

- Training failures caused by software defects are excluded only with logged justification and rerun under the same budget.
- Seeds are not removed because they produce poor performance.

- The full planned seed set should be completed before confirmatory interpretation unless a documented resource limit forces termination.
- Protocol amendments are dated and separated from post-hoc exploratory analyses.

9. Claim boundary

This protocol can support claims about the implemented task family and training setup. It cannot by itself establish general laws of swarm intelligence, biological collective behavior, physical-robot performance, communication thresholds, or universal phase transitions.

Figures

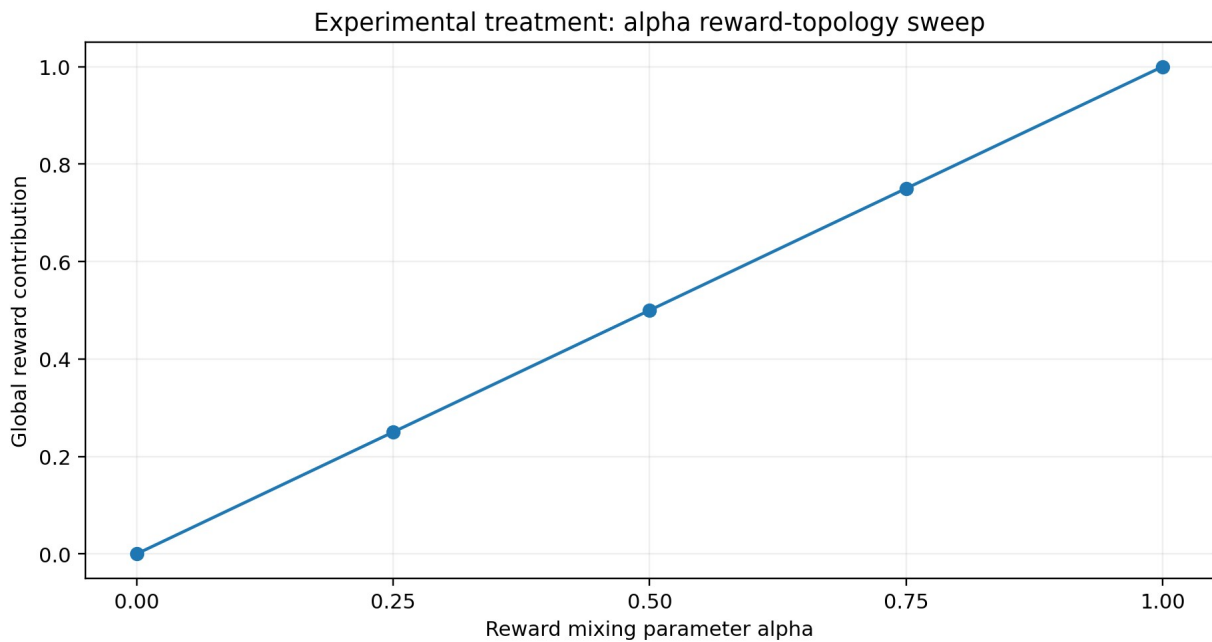


Figure 1. Pre-specified alpha treatment grid for the learned-policy experiment.

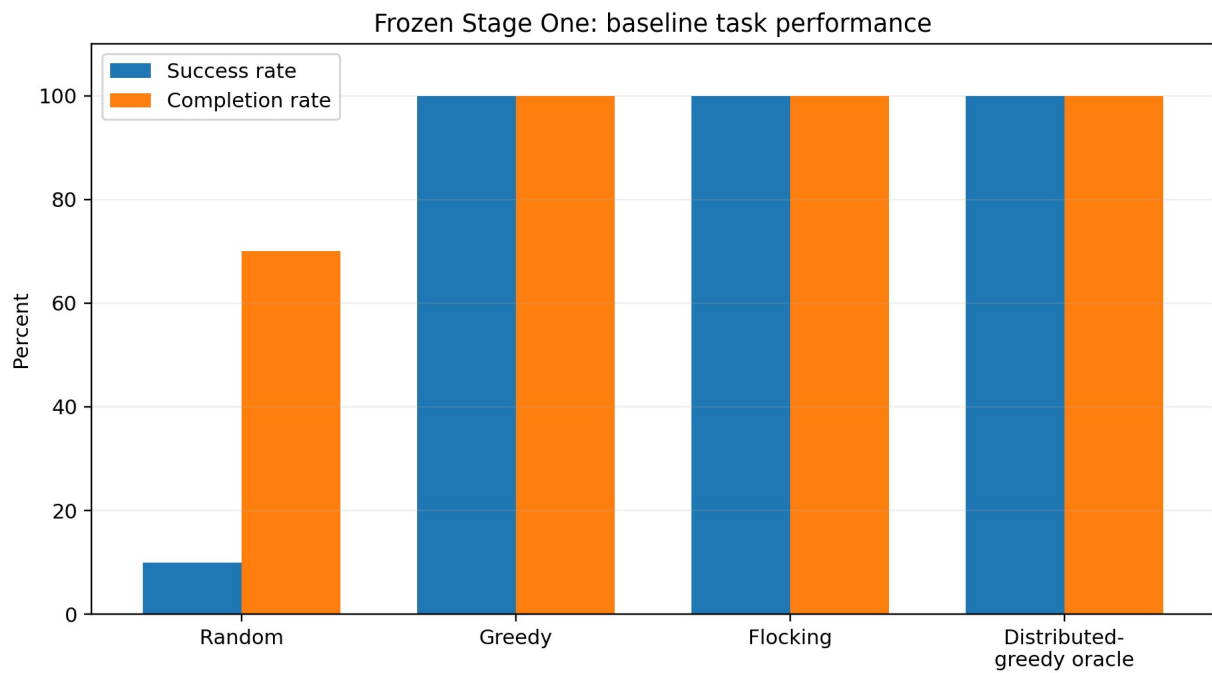


Figure 2. Frozen heuristic/control reference performance that the learned-policy experiment will compare against.